

Zhi Cheng Wong

Lead **Developer**/Technical Architect

2700 Grand Ave S
Minneapolis, MN 55408
Phone: 612-564-8559
E-mail: zhichengwong1@gmail.com

I'm a senior **software developer** with extensive experience including **leadership** and **architecture** experience. I am looking for a senior position that will make best use of my skills and expertise. I am a results-oriented **developer** with extensive experience working at all stages of the full life cycle of **software** projects. I have worked on teams large and small with other IT professionals including business analysts, architects, quality assurance staff, stake holders, and end users. I possess **good** communication skills as well as strong technical problem-solving skills. I have a strong work ethic and a drive to perform.

Technical Skills

Primary Skills:

C#, Javascript, Typescript, HTML, CSS

Languages:

C#, C/C++, CSS, HTML, Python, **Java**, JavaScript, PHP, SQL, TypeScript, VB.Net

Technologies:

Machine learning, n-tier **architecture**, XML & XSL, REST **web services**, SOA, SOAP **web services**, AI, Neural Networks, Large Language Models, Langchain, Vector Databases

Databases:

Microsoft SQL Server, Oracle, MySQL, Firebase, Pinecone, Vector Databases, ChromaDB, GraphQL, Entity Framework

Frameworks:

.Net, .Net Core, ASP.Net, Angular 2/4/7/8, Entity Framework, Vue, WCF, Windows Forms

Libraries:

JQuery, NUnit

Tools:

Apache, Crystal Reports, Docker, Git, Linux, Node.js, Octopus Deploy, Selenium, Subversion, Team Foundation Server, Teamcity, TensorFlow, Visual Studio, Visual Studio Code, SSRS/SSIS

Cloud Providers:

Amazon Web Services (AWS), Microsoft Azure **Cloud**, **Google** Cloud, LiquidWeb

Professional Experience

Senior Developer

US Bank (through Dexian)
Minneapolis, MN
8/2024-5/2025

At US Bank, I was a consultant that worked with the mortgage underwriting quality assurance group on a project to modernize and migrate a legacy .NET Framework ETL system to Azure cloud-based .NET Core.

Notable Accomplishments:

- Refactored three ETL programs and migrated them over to use .net core on a cloud platform.
- Designed a modular architecture with replaceable, class-based components to simplify future updates and accommodate new data transformation logic, reducing development effort for future ETL pipelines.
- Collaborated with cross-functional teams (QA, business analysts) to ensure transformed data met regulatory and business requirements for loan underwriting risk assessment.
- Mentor more junior developers on problem solving and explaining the new architecture.
- Aid in troubleshooting production problems that arose.
- Aid in solving deployment problems.

Lead Developer

Doubleday and Cartwright Co.
New York, New York (remote)
5/2022-4/2024

At Doubleday and Cartwright, I was part of small group that moved quickly to develop new products and implement customer projects. The technologies I worked with the most were HTML, Javascript, Typescript, MySQL, and Google Cloud services.

Notable Accomplishments:

- Developed an app to find qualified personnel on LinkedIn using a ChatGpt large language model and Langchain.
- Large Language Model GPU assisted fine tuning
- Developed several new ecommerce sites.
- Migrated a client's web site and data backend to AWS and maintained operations on that site.
- Reduced AWS cloud costs for one customer by more than 50%.
- Developed microservices for working with cryptocurrency blockchains and NFT minting.

Technical Architect

Cognizant (placed at 3M Corp)
Maplewood, Minnesota
12/2020-5/2022

At Cognizant, I was part of a large group that worked on a manufacturing resource planning (MRP) related **software** system. Our **software** was used by many of 3M's manufacturing plants around the world to manage their manufacturing workloads.

My own work was to review work done by developers, participate in **software** deployments, and implement **software** infrastructure related projects. I also participated in **architecture** planning and review sessions as well as troubleshooting sessions.

The technologies we relied upon most were **web**-based apps, C#, html, JavaScript, and SQL.

Notable Accomplishments:

- Designing and implementing new **services** to support single sign on across multiple enterprise applications.
- Troubleshoot and update our error logging system.
- Contribute to a migration from on premise IT infrastructure to **cloud** based AWS **services**.
- Update Bootstrap versions.

Senior Developer

Optum Health
Eden Prairie, Minnesota
08/2020-10/2020

At Optum, I worked on implementing a new application to aid in accounting for ongoing expenses. This application was a **web**-based app that utilized Angular 8 for the front end and used a Node.js for the backend.

Notable Accomplishments:

- In charge of implementing the Sequelize ORM product in our **software** product.
- Upgrade Angular versions.

Lead Developer

Strategic Education Incorporated (through Multivision Inc)
Minneapolis, Minnesota
12/2017-4/2020

1. At this workplace, I was the lead **developer** working in an agile project environment and work closely with other developers, scrum masters, QA staff, and dev ops. It was a fast-paced work environment with a focus on sprint velocity. I provided

mentoring to the more junior developers on the team and also assigned them work tasks.

2. The work in my group revolved around **processing** new student leads. We would filter and score new leads to find the most promising students for the university.
3. Our code was built using modern best practices including unit test and automated testing. Code was integrated using CI and CD tools. We worked closely with dev ops for deployments.
4. My work involved several websites and about a dozen back end programs to handle business processes. Technologies used include Asp.net MVC, **Java** Servlets, REST APIs, C#, .Net, .Net Core, **Java**, SQL, HTML, CSS, JavaScript, TypeScript, and Angular 7/8.
5. Notable Accomplishments:
 6. Implementing new features to the existing system.
 7. Maintain day-to-day operations of the system.
 8. Analyze and resolve problems reported by the business involving lead **processing**.
 9. **Design** and implement new **web** pages and **web** service endpoints using ASP.Net MVC and WebAPI.
 10. Implement new **web** pages in Angular for staff members to refer potential students for admission.
 11. Implementation of new **software** to score **customer** leads.
 12. Development of neural network AI based scoring method for new **customer** leads.
 13. Analyze and optimize database performance on the four databases used by our group.

Senior Developer

Volt Technical Recruiting (stationed at Emerson Electric Company)

Chanhassen, Minnesota

11/2016-12/2017

At Emerson I worked on applications that configure manufacturing workstations. I worked in a small team and worked closely with my team members and supervisor. I was the sole **developer** working on two applications developed utilizing the Angular 2/4 framework to provide a responsive interface to users located around the world at Emerson factories.

Notable Accomplishments:

- Development and testing of a new Angular application to set up label printing workstations at multiple factories.
- Development and testing of a new Angular application to create and edit labels for printing stations.

Senior Developer

Comprehensive Pharmacy Services (through Signature Consulting)
Brooklyn Park, Minnesota
9/2016-11/2016

At Comprehensive Pharmacy Services I worked on a project to integrate a newly acquired web-based software product with the existing portfolio of applications and software services. I also worked on financial reports using SSRS with an ASP MVC front end. Kendo UI components were used on this site.

Notable Accomplishments:

- Work on new website to manage pharmacies
- Developed new SSRS report to analyze site profitability.

Senior Developer

Conklin Company, Inc. (through Insight Global)
Shakopee, Minnesota
4/2014-4/2016

At Conklin I worked closely with a small team to develop new middleware layers during the transition to a new MRP platform. The environment was tight knight and fast paced with and I collaborated closely with team members and supervisors. The work consisted of WinForms development in C# and also SQL development.

Notable Accomplishments:

- Developed new middleware layer to transfer sales and customer data between the new MRP system and existing legacy systems.
- Participated in the design and implementation of a new database to modernize and replace the aging prior system.
- Developed a sales trend reporting and forecasting application for the business. I worked with business stakeholders to gathering business requirements and complete testing. I also wrote functional and technical specification documents.
- Developed a new ETL application to process sales commissions. Implemented innovative techniques to reduce execution time 30-fold compared to the old process.
- Performed numerous updates to the order processing software in the legacy system to implement new features and fix bugs.
- Made updates to the legacy database schema and stored procedures to accommodate new MRP data and new business processes.
- Oversaw daily order processing. I worked to resolve any issues that arose from problematic orders.

Senior Developer

Thomson Reuters (through McKinley Consulting)

Eagan, Minnesota

9/2013-1/2014

At Thomson I worked in a team to develop a new client facing application for lawyers to communicate with their clients. The project was run under an agile development process and we utilized many development best practices including unit tests and code reviews with every check in. Technologies used were ASP.Net MVC, JavaScript, C#, Underscore.js, REST JSON **web services**.

Notable Accomplishments:

- Developed widget to provide new users a tour of the application.
- Developed code to create and delete chat messages between lawyers and their clients.
- Developed new user tour of the application.
- Wrote and tested against unit tests and **integration** tests.

Senior Developer

LabCorp (formerly Medtox)

New Brighton, Minnesota

9/2009-9/2013

At LabCorp, I was the lead **developer** and worked to develop new public facing **web** applications, refactored existing **web** applications, worked heavily on backend development and supported daily business operations. I advised on new technologies and lead initiatives to improve the development methodologies used. The **systems** at LabCorp's St. Paul location utilized an n-tier, service-oriented **architecture**.

Accomplishments include:

- Large, long term project in Asp.net/C# to refactor and re-architect all .net based **software** used by LabCorp's Minnesota location. This involved a new database schema and updating all apps to .net 4.0. There were 500,000+ lines of code in aggregate in the code base and 50000+ lines of html/aspx markup.
- **Design** and implementation of new medical test and supply ordering application which utilized Ajax to improve responsiveness and incorporated Bing maps to aid in ordering.
- Sole **developer** working on an application to catalog the possible cross reactions between drugs and LabCorp's drug testing products. This was an asp.net **web** based application coded in C# with a SQL Server backend storage schema designed and developed by myself. I also did all the CSS work for this site. With this project, I went through the entire **software** development lifecycle from

gathering specifications from the end users to the **design**, development, testing, deployment, and maintenance.

- Development of new WCF **web** service for PDF creation and document **processing**.
- **Design** and implement a new **web** site page for submitting specimen collections.
- **Design** and implementation of an order creation tool (to aid QA testers) that was capable of creating orders in various permutations and also importing **data** from Excel.
- **Design** and implement a **web** based tool to track the history of an order and any errors that may have occurred.
- Development of middleware to interface an automated telephone call system to the LabCorp database.
- Development of a console application that processed new medical test orders placed via ftp by clients.
- **Design** and implementation of an error logging tool that relied on **web services** to asynchronously log errors.
- Monitor backend operations for errors and resolve any problems that arise on a daily basis.
- Daily support duties for issues escalated from LabCorp client support team and also on call support.
- Modifications to SQL Server stored procedures to add new features and implement bug fixes.
- Wrote and used unit testing to get **good** code coverage. Wrote and ran automated functional tests.
- Performance tuning and troubleshooting for public facing websites
- Wrote and amended technical and functional specification documents for projects worked on.

Operations Analyst

Well Fargo

Minneapolis, Minnesota

7/2009-9/2009

At the Wells Fargo Home Mortgage division, I developed **web** and Windows forms applications to support loan processors and underwriters. My work involves requirements analysis, **software** architecting, database **design**, and testing of new projects throughout the **software** development lifecycle.

Notable Accomplishments:

- Gathered requirements from end users to **design** and develop a loan underwriter work scheduling, tracking, and productivity monitoring website. I also did the CSS work on this also.
- **Design** and implementation of database schema for the above **web** site.

- Utilize open source tools to create and customize a CRUD layer for the underwriter **web** site I developed.
- **Design** and implementation of a Windows forms application for reporting underwriter profitability.
- Wrote and modified SQL Server stored procedures to support the applications I developed.

Developer/Analyst

Hallmark Insights Co.
Minneapolis, Minnesota
9/2007-05/2009

In this position I worked on Hallmark Insights' public facing **web** ordering site, internal business **software**, and provided support for the company's order **processing systems**. My work focused on supporting the company's order management system which included credit card **processing** functionality. Besides this, I also worked on numerous projects for Hallmark Insights' clients in order to implement product customizations for them and to accommodate their other special needs.

Accomplishments include:

- Primary responsibility for the company's internal order management system.
- Conversion of a major ASP.Net 1.1 application to version 2.0 and later to version 3.5.
- **Design** and implement changes to credit card **processing** of orders in order to comply with Sarbanes-Oxley regulations.
- Performance tuning of Oracle queries.
- Assisted, identified and provided solutions for critical production issues to ensure business process continuity.

Web Developer

Ideas to Go Co.
Minneapolis, Minnesota
10/2005-8/2007

I worked on internal as well as public facing **web** applications involving **Web 2.0** technologies. My team was small team and I was involved in a lot of the **software design** decisions and architecting. I was also involved with requirements gathering which involved working with end users and internal stake holders.

The applications I worked on were implemented using ASP.Net 2.0, VB.Net, and small amounts of C#. Microsoft SQL Server 2000/2005 served as the database back end and I also have written SQL queries, stored procedures, and made database **design** decisions.

Accomplishments include:

- A blogging site which I had a wide degree of control over the features and implementation of.
- A public facing idea collection website for PetSafe which collected pet product ideas from the general public and allowed for the public to vote on these ideas in order to find the “best” idea.
- A stenography application. This application collects and displays ideas from multiple users simultaneously. AJAX.Net, javascript, responsive web design, web services, and custom server controls were used to provide greater interactivity. The application included features such as user accounts, account management, multiple projects, idea collection, idea scoring, and report generation.
- Developed asp.net custom server controls to score contest entries.
- Worked on the build out of a new regional office where I configured servers and set up network connections for that location.

Help Desk Level III

Unisys

St Louis Park, Minnesota

5/2005-10/2005

In this position, I worked on a contract Unisys had with Dell to provide support to Park Nicollet Health Services. The facility I worked at was a six building campus where I was one of approximately four level 2 technicians. In this position, I helped clients with their software and hardware problems. I also performed desktop & laptop repairs. This position required a lot of individual initiative to take and resolve work tickets and also great multitasking skills in order to stay on top of an erratic and eclectic work load.

User Administrator

Cargill

Eden Prairie, Minnesota

6/2000-12/2000

At Cargill, I administrated user access rights and privileges for employees.

- Help onboard new hires by setting up user accounts and necessary system access.
- Configure and manage access privileges for employees to ensure they have the appropriate level of access to perform their job duties.
- Ensure compliance with company security policies and procedures regarding user access management.

Quality Assurance Specialist Intern

Fourth Shift Corporation

Bloomington, Minnesota
1996-1998

As a member of the QA team, I had sole responsibility over Fourth Shift's collection of automated **software** tests.

Accomplishments include:

- Designed, managed, and ran automated **software** testing scripts using Rational SQA Suite including Rational Robot.
- Develop database migration **data** integrity verification **software** using Access.
- Hands on **software** testing.

Undergraduate Research Assistant

University of Minnesota, Mechanical **Engineering** Department
08/1996-05/1998

As a research assistant, I helped to run experiments aimed at fabricating novel new nanoscale materials with plasma deposition.

Accomplishments include:

- Programmed hardware device wrapper for **data** acquisition scientific equipment incompatible with LabView **software**.
- Programmed a computer controlled substrate temperature control program using Labview.
- Operated and maintained **data** acquisition computers.
- Performed the duties of a lab technician
- Made CAD illustrations of lab apparatuses using Pro Engineer.

Certifications

Amazon **AWS** **Cloud** Practitioner

Education

Bachelors of Computer Science
University of Minnesota Twin Cities
Institute of Technology
Focus on artificial **intelligence** and machine learning
Graduated in August 2003

Zhi Wong

- Minneapolis, MN, US

Contact Information

- 01c-c48-7ot@mail.dice.com
- 6125648559

Work History

Total Work Experience: 29 years

- Senior **Software** Engineer US Bank
Aug 01, 2024
- Lead **Software** Engineer Doubleday And Cartwright
May 01, 2022
- Technical Architect Cognizant (Placed At 3M Corp)
Dec 01, 2020
- Senior **Developer** Judge Technical **Services** Inc
Aug 01, 2020
- Lead **Developer** Strategic Education Incorporated
Dec 01, 2017
- Senior **Developer** Volt Technical Recruiting (Placed At Emerson Electric)
Nov 01, 2016
- Senior **Developer** Comprehensive Pharmacy **Services**
Sep 01, 2016
- Senior **Developer** Conklin Company, Inc
Apr 01, 2014

- **Senior Developer** Thomson Reuters
Sep 01, 2013
- **Senior Developer** Labcorp
Sep 01, 2009
- **Operations Analyst** Well Fargo
Jul 01, 2009
- **Developer/Analyst** Hallmark Insights
Sep 01, 2007
- **Web Developer** Ideas To Go
Oct 01, 2005
- **Help Desk Level III** Unisys
May 01, 2005
- **User Administrator** Cargill
Jun 01, 2000
- **Quality Assurance Specialist Intern** Fourth Shift Corporation
Jan 01, 1996
- **Undergraduate Research Assistant** University Of Minnesota
Jan 01, 1996

Education

- **Bachelors** | University of Minnesota Twin Cities

Skills

- **aws** - 3 years
- **php** - 2 years
- **azure** - 1 years
- **google cloud** - 1 years

- **machine learning** - 1 years
- **python** - 1 years
- **c#** - 21 years
- **.net** - 19 years
- **database** - 19 years
- **web services** - 19 years
- **css** - 17 years
- **html** - 17 years
- **sql** - 17 years
- **javascript** - 15 years
- **agile** - 13 years
- **java** - 13 years
- **mentorship** - 13 years
- **mvc** - 13 years
- **scrum** - 13 years
- **qa** - 19 years
- **reporting** - 15 years
- **typescript** - 4 years

Work Preferences

- Desired Work Settings: Remote or On-Site or Hybrid
- Likely to Switch: True
- Willing to Relocate: False
- Travel Preference: 0%
- Work Authorization:
 - US
- Work Documents:
 - US Citizen
 Security Clearance: False
- Third Party: False
- Employment Type:
 - Full-time
 - Contract - Independent
 - Contract - W2
 - Contract to Hire - Independent
 - Contract to Hire - W2

Profile Sources

- Dice: <https://www.dice.com/employer/talent/profile/29e9a3f77247b8de6010cfaf114c1d38>